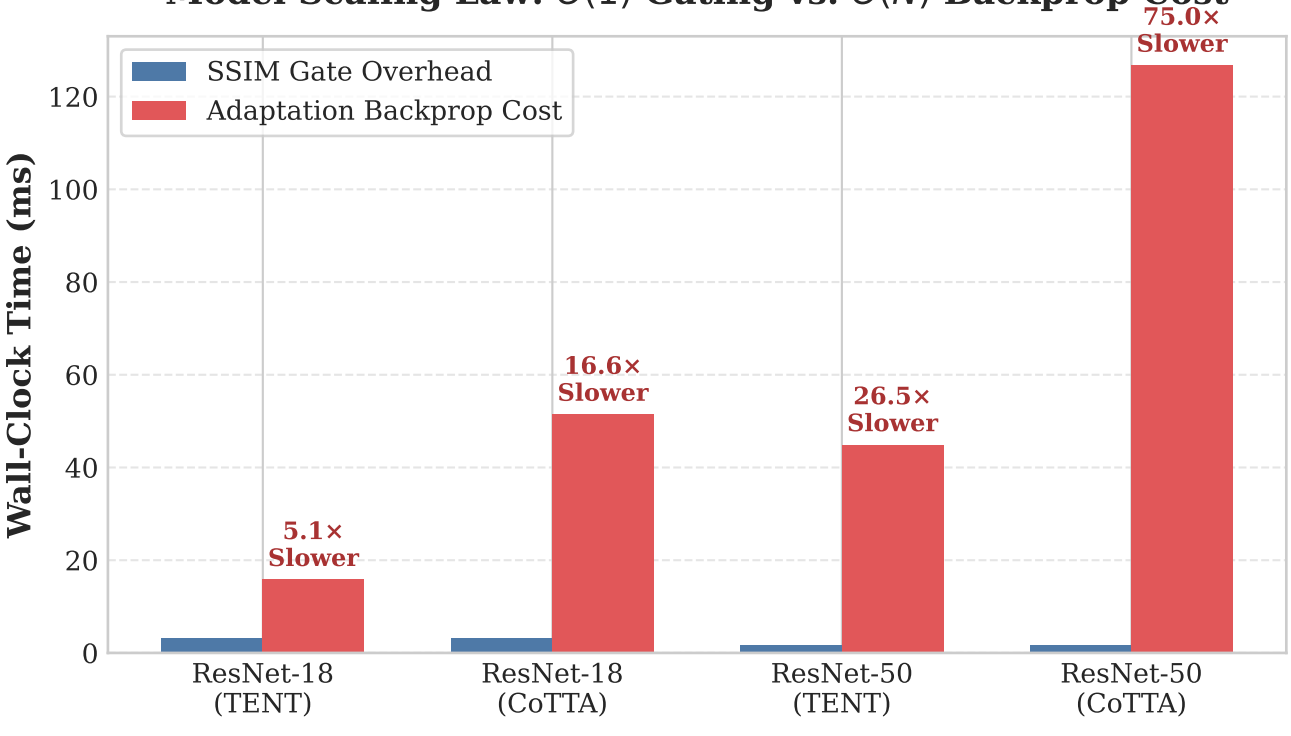
Model Scaling Law: $O(1)$ Gating vs. $O(N)$ Backprop Cost
SSIM Gate Overhead
Adaptation Backprop Cost
Wall-Clock Time (ms)
0
20
40
60
80
100
120
5.1× Slower
16.6× Slower
26.5× Slower
75.0× Slower
ResNet-18 (TENT)
ResNet-18 (CoTTA)
ResNet-50 (TENT)
ResNet-50 (CoTTA)